

Correlation and Regression

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What is correlation and correlation coefficient?

The relationship between two or more variable is called correlation. The primary objective of correlation analysis is to measure the strength or degree of relationship between two or more variables. If the change in one variable effects a change in the other variable, the variable are said to be correlated.

Method of studying simple correlation:-

1. Scatter Diagram method.
2. Karl Pearson's coefficient of correlation.
3. Spearman's Rank correlation.

Correlation coefficient:-

The mathematical measures of correlation are called correlation coefficient (r_{xy}/r).

$$r_{xy} = \frac{n \sum xy - \sum x \sum y}{\sqrt{\{n \sum x^2 - (\sum x)^2\} \{n \sum y^2 - (\sum y)^2\}}}$$

This formula was given by Karl Pearson's in 1890.

Assumptions of Pearson's correlation coefficient:-

1. There is a linear relationship between two variables when the two variables are plotted on a scatter diagram. a straight line will be formed by the points.
2. Cause and effect relation exists between different forces operating on the item of the two variable series.

consider the following data

x	y
1	6
2	4
3	3
4	5
5	4
6	2

Here,

$$\sum x = 21, \sum y = 24, \sum x^2 = 91, \sum y^2 = 106, \sum xy = 75, n = 6$$

We know,

$$r_{xy} = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sqrt{\left\{ \sum x^2 - \frac{(\sum x)^2}{n} \right\} \left\{ \sum y^2 - \frac{(\sum y)^2}{n} \right\}}}$$

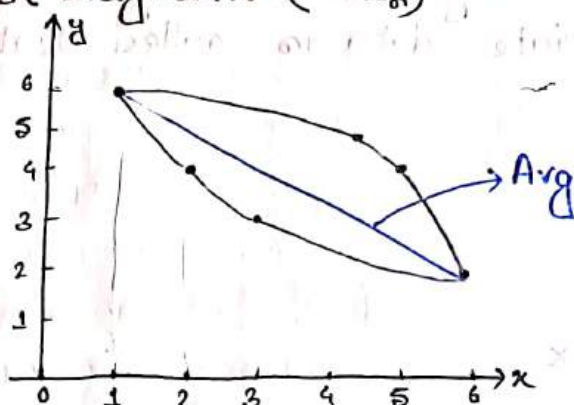
$$= \frac{6 \times 75 - 21 \times 24}{\sqrt{\{6 \times 91 - (21)^2\} \{6 \times 106 - (24)^2\}}}$$

$$= \frac{-54}{\sqrt{105 \times 60}}$$

$$= -0.680$$

(Ans)

Scatter Diagram:- (चिह्नित)



#Consider the following data:-

<u>x</u>	<u>y</u>
11	16
12	14
13	18
14	15
15	14
16	12

Here,

$$\sum x = 81, \sum y = 84, \sum x^2 = 1111, \sum y^2 = 1186, \sum xy = 1125.$$

$$n = 6$$

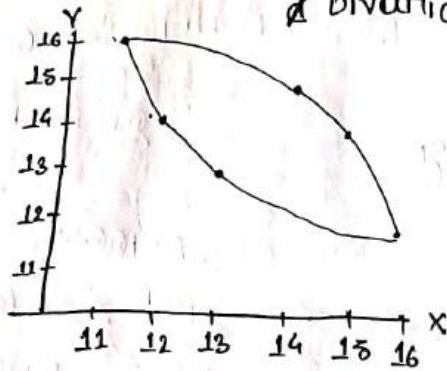
we know,

$$r_{xy} = \frac{n \sum xy - \sum x \sum y}{\sqrt{\{n \sum x^2 - (\sum x)^2\} \{n \sum y^2 - (\sum y)^2\}}}$$

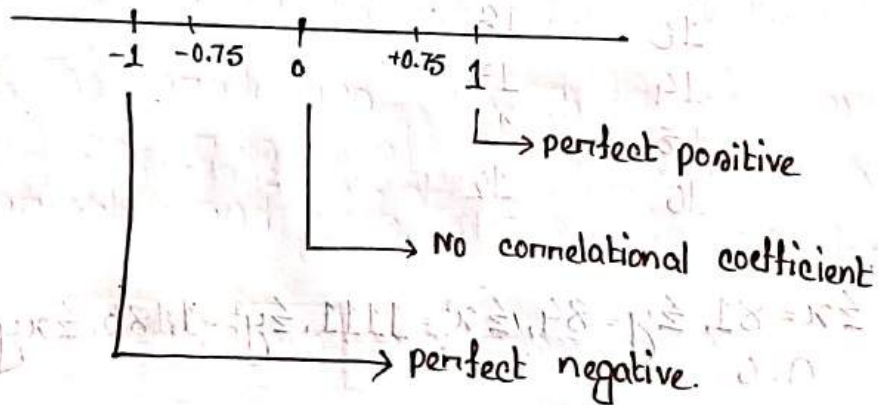
$$= \frac{6 \times 1125 - 81 \times 84}{\sqrt{\{6 \times 1111 - (81)^2\} \{6 \times 1186 - (84)^2\}}}$$

$$= \frac{-54}{\sqrt{105 \times 2460}} = -0.68 \quad (\text{Ans}).$$

Scatter Diagram: The diagrammatic way of representing bivariate data is called scatter diagram.



$$-1 \leq r \leq +1$$



- ① $r=0 \rightarrow$ x and y are independent. There is no correlation coefficient.
- ② $0 < r_{xy} < 0.75 \rightarrow$ Positive correlation coefficient.
- ③ $0.75 \leq r < 1 \rightarrow$ Strongly positive correlation coefficient.
- ④ $r=1 \rightarrow$ Perfect positive correlation coefficient.

Intercepts of r :

$r = +1 \rightarrow$ a perfect positive relationship between x and y . Symmetric data is

$r = -1 \rightarrow$ a perfect negative relationship between x and y .

$r = 0 \rightarrow$ there is no linear relationship between x and y . Here two variables are linearly independent.

$0 < r < 1 \rightarrow$ a positive relationship between x and y .

$-1 < r < 0 \rightarrow$ a negative relationship between x and y .

0.75 1

$\rightarrow$ perfect

$\rightarrow$ No connection

⑤ $-0.75 < r < 0 \longrightarrow$ negative correlation coefficient.

⑥ $-1 < r \leq -0.75 \longrightarrow$ Strongly negative correlation coefficient

⑦ $r = -1 \longrightarrow$ Perfectly negative correlation coefficient

Comment for the previous problem:-

Ans. $r_{xy} = 0.68$.

The correlation between x and y is negative.

☐ Properties of correlation coefficient:-

1. Correlation coefficient is independent of change of origin and scale measurement.

2. Correlation coefficient lies between -1 to $+1$.

i.e.

$$\boxed{-1 < r_{xy} < +1}$$

3. Correlation coefficient is symmetric,

i.e. $\boxed{r_{xy} = r_{yx}}$

4. Correlation coefficient is the geometric mean of regression coefficient

i.e. $\boxed{r_{xy} = \sqrt{b_{xy} \cdot b_{yx}}}$

5. For two independent variables correlation coefficient

is zero.

5. Correlation coefficient is unit free.

What is Regression and what is regression coefficient?

Regression is the ~~fundame~~ functional relationship between two variables and of the two variables one may represent cause and other may represent effect.

This term was used by a famous Biometrician Sir F. Galton in 1877.

Example:- The production of paddy of amount y is dependent on rainfall of amount x . Here x is independent variable and y is dependent variable.

$$y = a + bx + e \rightarrow \text{regression eqn}$$

Here,

y = Dependent variable / Predicted variable

a = constant / intercept term.

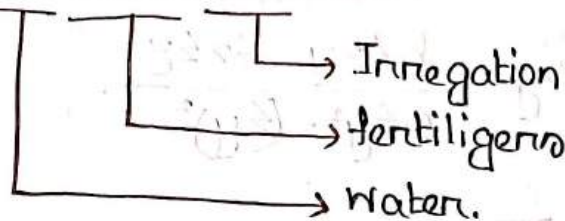
b = regression coefficient.

x = Independent variable / Predictor variable / regressor

e = error term.

The yield of paddy depends on water, fertilizers and irrigation etc.

$$Y = a + b_1x_1 + b_2x_2 + b_3x_3 + e$$



→ The mathematical measures of regression are called the coefficient of regression.

$$b_{yx} = \frac{n \sum xy - \sum x \cdot \sum y}{n \sum x^2 - (\sum x)^2}$$

→ Y = dependent
 x = Independent.

$$b_{xy} = \frac{n \sum xy - \sum x \cdot \sum y}{n \sum y^2 - (\sum y)^2}$$

→ x = dependent
 Y = Independent

Let (x_1, y_1) (x_2, y_2) ... (x_n, y_n) be the pairs of n observations, Then the regression coefficient y on x denoted by b_{yx} and regression coefficient x on y ,

Show that $\rho_{xy} = \sqrt{b_{yx} \cdot b_{xy}}$

We know,

$$b_{yx} = \frac{R_{zy}x - R_{xz}y}{n(x^2 - xz)}$$

$$b_{xy} = \frac{R_{zx}y - R_{yz}x}{n(y^2 - yz)}$$

$$\therefore \text{R.H.S} = \sqrt{b_{yx} \cdot b_{xy}}$$

$$= \sqrt{\frac{R_{zy}x - R_{xz}y}{n(x^2 - xz)} \cdot \frac{R_{zx}y - R_{yz}x}{n(y^2 - yz)}}$$

$$= \sqrt{\frac{(R_{zy}x - R_{xz}y)^2}{n(x^2 - xz) \{n(y^2 - yz)\}}}$$

$$= \frac{R_{zy}x - R_{xz}y}{\sqrt{n(x^2 - xz) \{n(y^2 - yz)\}}}$$

$$= \rho_{xy}$$

$$= \text{L.H.S.}$$

$$\therefore \rho_{xy} = \sqrt{b_{yx} \cdot b_{xy}} \quad (\text{shown})$$

☐ Properties of Regression:-

1. Regression coefficient is independent of change of origin but not of scale.

2. Regression coefficient lies between $-\infty$ to $+\infty$

i.e.
$$-\infty < b_{xy}/b_{yx} < +\infty$$

3. Regression isn't symmetric.

i.e.
$$b_{xy} \neq b_{yx}$$

4. ~~Reasi~~ The geometric mean of regression is equal to the correlation coefficient.

i.e.,
$$r_{xy} = \sqrt{b_{yx} \cdot b_{xy}}$$

5. The arithmetic mean of regression coefficient is greater than correlation coefficient.

i.e.
$$r_{xy} \leq \frac{b_{xy} + b_{yx}}{2}$$

6. Regression coefficient is not unit free.

☐ Difference between the regression coefficient & correlation coefficient:-

Correlation coefficient	Regression coefficient
1. It lies between -1 to +1.	1. It lies between $-\infty$ to $+\infty$.
2. It is symmetric. $r_{xy} = r_{yx}$	2. It is not symmetric $b_{xy} \neq b_{yx}$
3. For two independent variable correlation coefficient is zero.	3. -

Correlation coefficient	Regression coefficient
1. The numerical value by which we measure the strength of linear relationship between two or more variable is called correlation coefficient.	1. The mathematical measures of regression are called the coefficient of regression.
2. Correlation coefficient is independent of change of origin and scale of measurement.	2. Regression coefficient is independent of change of origin but not scale.
3. It lies between -1 to +1 $-1 \leq r_{xy} \leq +1$	3. It lies between $-\infty$ to $+\infty$ $-\infty < b_{xy} / b_{yx} < +\infty$

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4. It is symmetric.
 i.e. $r_{xy} = r_{yx}$.

4. It is not symmetric.
 i.e. $b_{yx} \neq b_{xy}$.

5. It is always unit free.

5. It is not pure number.

6. When $r=0$ then the variables are connected.

6. When $r=0$, then the two lines of regression are perpendicular to each other.

Regression Equation:-

The regression Equation of y on x is expressed as follows-

$$y = a + bx$$

$$a = y - bx$$

$$= \bar{y} - b\bar{x}$$

and $b_{yx} = \frac{n \sum xy - \sum x \sum y}{\{n \sum x^2 - (\sum x)^2\}}$

Similarly, the regression equation of x on y is expressed as follows:

$$x = a + by$$

$$\therefore a = x - by$$

$$= \bar{x} - b\bar{y}$$

and $b_{xy} = \frac{n \sum xy - \sum x \sum y}{\{n \sum y^2 - (\sum y)^2\}}$

y = Dependent variable

x = independent variable

a = intercept term

b = slope of the line.

$$a = \bar{y} - b\bar{x}$$

$$= \frac{\sum y}{n} - b \frac{\sum x}{n}$$

y = Independent variable

x = Dependent variable

a = intercept term

b = slope of the line

$$a = \bar{x} - b\bar{y}$$

$$= \frac{\sum x}{n} - b \frac{\sum y}{n}$$

Prob-1

Temp of water	68	65	70	62	60	55	58	65	69	63
Reduction pulse rate	2	5	1	10	9	13	10	3	4	6

Sol:-

x	y	x ²	y ²	xy
68	2	4624	4	136
65	5	4225	25	325
70	1	4900	1	70
62	10	3844	100	620
60	9	3600	81	540
55	13	3025	169	715
58	10	3364	100	580
65	3	4225	9	195
69	4	4761	16	276
63	6	3969	36	378

Here, $\sum x = 635$, $\sum y = 72$, $\sum x^2 = 40537$, $\sum y^2 = 541$, $\sum xy = 3835$

$\therefore$ Correlation coefficient,

$$[n=10]$$

$$r_{xy} = \frac{n \sum xy - \sum x \sum y}{\sqrt{\{n \sum x^2 - (\sum x)^2\} \{n \sum y^2 - (\sum y)^2\}}}$$

$$= \frac{10 \cdot 3835 - 635 \times 72}{\sqrt{\{10 \cdot 40537 - (635)^2\} \{10 \cdot 541 - (72)^2\}}}$$

$= -0.94$. (Ans)

The result -0.94 indicated that the correlation coefficient between temperature of water and reduction in pulse rate is highly negatively connected.

Prob-2

x	1	2	3	4	5	6
y	6	4	3	5	4	2

- Find r_{xy} .
- Draw Scatter Diagram.
- Estimate the value of y when $x=3$.
- Calculate regression line of x on y .
- Predict the value of x when $y=4.5$.

Sol:-

iii. $\sum x = 21$, $\sum y = 24$, $\sum x^2 = 91$, $\sum y^2 = 106$, $\sum xy = 75$

the regression eqⁿ of y on x is

$$y = a + bx \quad \text{--- (i)}$$

Here,

$$a = \bar{y} - b\bar{x}$$

$$= \frac{\sum y}{n} - b \frac{\sum x}{n}$$

$$= \frac{24}{6} - b \frac{21}{6} \quad [n=6] \quad \text{--- (ii)}$$

and,

$$b_{\hat{y}\hat{x}} = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$$

$$= \frac{6 \times 75 - 21 \times 24}{6 \times 91 - (21)^2}$$

$$= -0.514,$$

Putting the value of b on eqn (ii),

$$a = \frac{24}{6} - (-0.514) \frac{21}{6}$$

$$= 5.8.$$

Now, putting the value of a and b in eqn (i),

$$y = 5.8 + (-0.514)x$$

$$y = 5.8 - 0.514 \times 3 \quad [\because x = 3]$$

$$= 4.258. \text{ (Ans),}$$

Fundamentals of Probability

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Experiment:- Experiment is an act that can be repeated under given conditions.

Ex- Tossing a coin, throwing a dice etc.

Outcomes:- The result of an experiment are called outcomes.

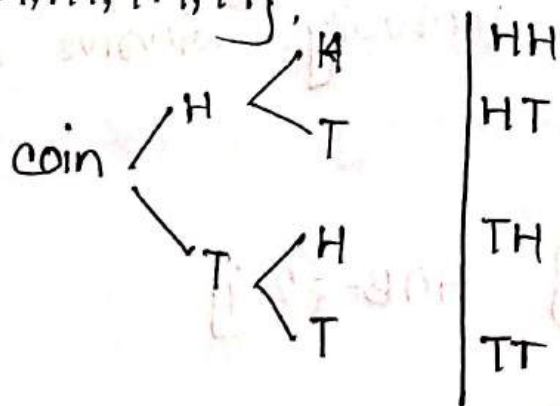
Random experiment:- Experiment are called random experiments if the outcomes depend on chances and cannot be predicted with certainty.

Ex- Tossing a fair coin, throwing of dice etc.

Sample Space:- The collection or totality of all possible outcomes of a random experiment is called sample space. It is denoted by S or Ω

Example:- 1. If we toss a coin the sample space is $S = \{H, T\}$.
2. If we toss a coin two times, the sample space is.

$S = \{HH, HT, TH, TT\}$



Each element of a sample space is called sample point.

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Event:- Any subset of a sample space is called event.

It is denoted by A, B, C . There are two types of event.

They are:-

1. Simple Event.
2. Compound Event.

Mutually Exclusive Event:- Two events are said to be mutually exclusive if they have no common points.

Ex- $A = \{3, 5, 1\}$ $B = \{2, 4, 6\}$ then $A \cap B = \emptyset$

$$AB = \emptyset \quad P(AB) = P(A \cap B) = 0.$$

If two events are mutually exclusive, then their probability is 0.

Non-mutually Exclusive Event:- Two events are said to be not mutually exclusive event if they have common points.

If A and B are not mutually exclusive events, then

$$AB \neq \emptyset$$

$$A = \{1, 2, 4\} \quad B = \{2, 4, 6\} \quad A \cap B = \{2, 4\}$$

Conditional Probability:- If A and B are two events in S. Then the conditional probability of A for given value of B denoted by $P(A|B)$ is defined by

$$P(A|B) = \frac{P(AB)}{P(B)} ; P(B) > 0$$

Similarly, $\rightarrow$ A conditional B / A given B, B ঘটনা আগে ঘটে, B কে base করে A নির্ণয়

$$P(B|A) = \frac{P(AB)}{P(A)} ; P(A) > 0$$

$\rightarrow$ B conditional A / B given A, A ঘটনা আগে ঘটে, A কে base করে B নির্ণয়

Independent Event:- Two events A and B are said to be independent, if and only if one of the following conditions holds:

$$i) P[AB] = P[A]P[B]$$

$$ii) P[A|B] = P[A]$$

$$iii) P[B|A] = P[B]$$

Dependent Event:- Two events A and B are said to be dependent if and only if one of the following conditions holds:

$$i) P[AB] \neq P[A]P[B]$$

$$ii) P[A|B] \neq P[A]$$

$$iii) P[B|A] \neq P[B]$$

Definition of Probability:-Classical or Mathematical Probability (Priori Probability):-

If A is the event in S then the probability of the event A is denoted by $P(A)$ is defined as

$$P(A) = \frac{\text{favourable number of outcomes}}{\text{total number of outcomes}}$$

$$P(A) = \frac{N(A)}{N(S)} = \frac{m}{n}; \quad 0 \leq P(A) \leq 1$$

Example:-

$$S = \{1, 2, 3, 4, 5, 6\} \quad A = \{1, 2\}$$

$$N(S) = 6$$

$$N(A) = 2$$

$$\therefore P(A) = \frac{N(A)}{N(S)} = \frac{2}{6} = \frac{1}{3}$$

Axiomatic Probability:-

Suppose S is a sample space and A is an event of this sample space. Then the probability of the event A denoted by $P(A)$ must satisfy the following axioms:

- i. $P(A) \geq 0$
- ii. $P(S) = 1$
- iii. If A and B are mutually exclusive events.
- iv. $P(A_1 \cup A_2 \cup \dots \cup A_k) = P(A_1) + P(A_2) + \dots + P(A_k)$

Additive laws of Probability:

1. Mutually exclusive event
2. Non mutually exclusive event

If 'A' and 'B' are two events then,

$$P[A \cup B] = P[A] + P[B] \longrightarrow \text{mutually exclusive event}$$

$$P[A \cup B] = P[A] + P[B] - P[AB] \longrightarrow \text{non mutually exclusive event}$$

Theorem-1

The values of probability lies between 0 to 1.
 i.e. $0 \leq P(A) \leq 1$

Theorem-2

$$P(A) + P(\bar{A}) = 1$$

1. What are the different approaches of defining probability?

There are 4 approaches of defining probability.

They are —

1. Classical or mathematical definition of probability.
2. Empirical or statistical Probability.
3. Subjective Probability
4. Axiomatic Probability.

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2. Difference between experiment and random experiment

Experiment	Random Experiment
1. Experiment is an act that can be repeated under given conditions.	1. Experiment is called random experiment if its outcomes depends on chances and can't be predicted with certainty.
2. Its outcomes can be biased.	2. Its outcomes can't be biased.
3. An experiment can have any number of possible outcomes.	3. A random experiment has more than one possible outcomes.
4. Tossing with a biased coin.	4. Tossing a fair coin.

3. Difference between independent event and dependent event

Independent Event	Dependent Event
1. When probability of one event does not affect the probability of another event, they are known as independent events.	1. The probability of one event does affect the probability of another event, then they are called dependent events.

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4. Difference between mutually exclusive and non mutually exclusive event.

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5. Discuss classical probability and axiomatic probability

Ques-

In a factory, machines A, B, and C manufacture respectively 25%, 35%, 40% of their product of the total of their output 5, 4 and 2 percent are defective bolts respectively. A bolt is drawn at random from the product and is found to be defective. What are the probabilities of that it was manufactured by machine (i) A and (ii) C.

Let,

B_1 item is produced by machine A

B_2 " " " " " B

B_3 " " " " " C

D defective " " " " all machines.

We have,

$$P[B_1] = 25\% = 0.25$$

$$P[B_2] = 35\% = 0.35$$

$$P[B_3] = 40\% = 0.4$$

and,

$$P[D|B_1] = 5\% = 0.05$$

$$P[D|B_2] = 4\% = 0.04$$

$$P[D|B_3] = 2\% = 0.02$$

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☐ Sat ☐ Sun ☐ Mon ☐ Tue ☐ Wed ☐ Thu ☐ FriNow, we have find $P(B_1|D)$ and $P(B_3|D)$

We know,

$$P(B_i|D) = \frac{P(B_i) \cdot P(D|B_i)}{\sum_{i=1}^n P(B_i) P(D|B_i)}$$

$$\therefore P(B_1|D) = \frac{P(B_1) \cdot P(D|B_1)}{P(B_1) P(D|B_1) + P(B_2) P(D|B_2) + P(B_3) P(D|B_3)}$$

$$= \frac{0.25 \times 0.05}{0.25 \times 0.05 + 0.35 \times 0.04 + 0.4 \times 0.02}$$

$$= \frac{0.0125}{0.0345}$$

$$= 0.323. \text{ (Ans.)}$$

$$\therefore P(B_3|D) = \frac{P(B_3) \cdot P(D|B_3)}{\sum P(B_i) P(D|B_i)}$$

$$= \frac{0.4 \times 0.02}{0.0345}$$

$$= 0.232. \text{ (Ans.)}$$

State Additive law of Probability and multiplicative law of probability.

Non mutually exclusive event,

$$P(A \cup B) = P(A) + P(B) - P(AB)$$

Mutually exclusive event,

$$P(A \cup B) = P(A) + P(B)$$

If A, B and C are events then,

Non mutually exclusive event,

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(AB) - P(BC) - P(CA) + P(ABC)$$

Mutually exclusive event,

$$P(A \cup B \cup C) = P(A) + P(B) + P(C)$$

Multiplicative law:-

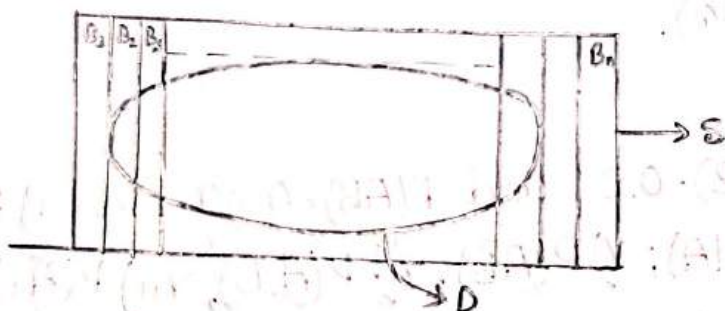
$$\begin{aligned} P(AB) &= P(A|B) \cdot P(B) \\ &= P(B|A) \cdot P(A) \end{aligned}$$

State Bayes theorem.

Let $B_1, B_2, B_3, \dots, B_n$ be a sequence of mutually exclusive events D be the event in S .

Then,

$$P(B_i|D) = \frac{P(B_i|D) \cdot P(B_i)}{\sum_{i=1}^n P(B_i) \cdot P(D|B_i)} ; i = 1, 2, 3, \dots, n.$$



Prblm-1

A and B are two mutually exclusive events with $P[A] = 0.35$ and $P[B] = 0.15$. Find $P[A \cup B]$, $P[\bar{A}]$, $P(A \cap B)$, $P(\bar{A} \cup \bar{B})$.

Given,

$$P[A] = 0.35,$$

$$P[B] = 0.15$$

$$\begin{aligned} \therefore P[A \cup B] &= P[A] + P[B] \\ &= 0.35 + 0.15 \\ &= 0.5, \end{aligned}$$

$$\begin{aligned} \therefore P(\bar{A}) &= 1 - P[A] \\ &= 1 - 0.35 \\ &= 0.65. \end{aligned}$$

$$\therefore P(A \cap B) = 0$$

Since, A and B are mutually exclusive.

$$\begin{aligned} \therefore P(\bar{A} \cup \bar{B}) &= P(\overline{A \cap B}) \\ &= 1 - P(A \cap B) \\ &= 1 - 0 \\ &= 1. \quad (\text{Ans}) \end{aligned}$$

Prblm-2

If $P(A) = 0.6$, $P(B) = 0.8$ and $P(AB) = 0.50$. Find i) $P(\bar{A})$, ii) $P(A \cup B)$, iii) $P(A|B)$, iv) $P(B|A)$; v) $P(A\bar{B})$, vi) $P(\bar{A} \cdot B)$, vii) $P(\bar{A}\bar{B})$, viii) $P(\bar{A} \cup \bar{B})$, xi) Are the events A and B independent? x) Are A and B mutually exclusive?

Given that,

$$P(A) = 0.6$$

$$P(B) = 0.8$$

$$P(AB) = 0.50$$

i. We know,

$$P(A) + P(\bar{A}) = 1$$

$$\therefore P(\bar{A}) = 1 - P(A)$$

$$= 1 - 0.6$$

$$= 0.4. \quad (\text{Ans})$$

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ii) By additive law of probability,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= 0.6 + 0.8 - 0.50$$

$$= 0.9 \text{ (Ans)}$$

$$\text{iii) } P(A|B) = \frac{P(AB)}{P(B)} \quad [P(B) > 0]$$

$$= \frac{0.50}{0.8}$$

$$= 0.625 \text{ (Ans)}$$

$$\text{iv) } P(B|A) = \frac{P(AB)}{P(A)}$$

$$= \frac{0.50}{0.6}$$

$$= 0.833 \text{ (Ans)}$$

$$\text{v) } P(A\bar{B}) = P(A \cap \bar{B})$$

$$= P(A) - P(AB)$$

$$= 0.6 - 0.50$$

$$= 0.10 \text{ (Ans)}$$

$$\text{vi) } P(\bar{A}B) = P(B) - P(AB)$$

$$= 0.8 - 0.50$$

$$= 0.30 \text{ (Ans)}$$

$$\text{vii) } P(\bar{A}\bar{B}) = P(\bar{A} \cap \bar{B})$$

$$= 1 - P(A \cup B)$$

$$= 1 - P(A \cup B)$$

$$= 1 - 0.9$$

$$= 0.1 \text{ (Ans)}$$

$$\text{viii) } P(\overline{A \cup B}) = 1 - P(A \cup B)$$

$$= 1 - 0.9$$

$$= 0.1$$

$$\text{ (Ans) }$$

ix) The events A and B are independent if

$$P[AB] = P[A]P[B]$$

Here,

$$P[AB] = 0.5.$$

$$\therefore P[A]P[B] = 0.6 \times 0.8 \\ = 0.48.$$

Ans, $P(AB) \neq P(A)P(B)$, the events are not independent.
(Ans)

x) A and B are mutually exclusive if $P[AB] = 0$

But,

$$\text{here } P(AB) = 0.5$$

which isn't equal 0.

So, A and B are not mutually exclusive event.

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Problem-3

A and B are two mutually exclusive events with $P[A] = 0.35$ and $P[B] = 0.15$. Find

i) $P[A \cup B]$

Given that,

$$P[A] = 0.35$$

$$P[B] = 0.15$$

$$\begin{aligned} \therefore P[A \cup B] &= P[A] + P[B] \\ &= 0.35 + 0.15 \\ &= 0.5. \text{ (Ans)} \end{aligned}$$

ii) $P(\bar{A})$.

We know,

$$P(A) + P(\bar{A}) = 1.$$

$$\begin{aligned} \Rightarrow P(\bar{A}) &= 1 - P(A) \\ &= 1 - 0.35 \\ &= 0.65. \text{ (Ans)} \end{aligned}$$

iii) $P(A \cap B)$.

$$P(A \cap B) = 0. \text{ (Ans)}$$

Are they mutually exclusive events.

$$\begin{aligned} \text{iv) } P(\bar{A} \cap \bar{B}) &= P(\overline{A \cap B}) \\ &= 1 - P(A \cap B) \\ &= 1 - 0 \\ &= 1. \text{ (Ans)} \end{aligned}$$

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Problem-4

If $P(A) = \frac{1}{3}$, $P(B) = \frac{3}{4}$, $P(A \cup B) = \frac{11}{12}$. Find i) $P(A|B)$ ii) $P(\bar{A}|B)$

Given that,

$$P(A) = \frac{1}{3}$$

$$P(B) = \frac{3}{4}$$

$$P(A \cup B) = \frac{11}{12}$$

We know,

$$P(A \cup B) = P(A) + P(B) - P(AB)$$

$$\Rightarrow \frac{11}{12} = \frac{1}{3} + \frac{3}{4} - P(AB)$$

$$\Rightarrow P(AB) = \frac{11}{12} + \frac{1}{3} + \frac{3}{4}$$
$$= \frac{1}{6}$$

$$\text{i) } P(A|B) = \frac{P(AB)}{P(B)}$$
$$= \frac{1/6}{3/4}$$

$$= \frac{2}{9} \text{ (Ans)}$$

$$\text{ii) } P(\bar{A}|B) = \frac{P(\bar{A}B)}{P(B)}$$
$$= \frac{P(B) - P(AB)}{P(B)}$$

$$= \frac{3/4 - 1/6}{3/4} = \frac{7}{9} \text{ (Ans)}$$

Random variable & Mathematical Expression

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Random variable:-

A random variable is a real valued function whose values are determined with the outcomes of a random experiment. It is usually denoted by X, Y, Z .

Let us consider the experiment of tossing two fair coins. The sample space of the experiment is,

$$S = \{HH, HT, TH, TT\}$$

Let X denotes the number of heads,

So,

<u>Sample Point</u>	<u>Number of Head</u>
HH	2
HT	1
TH	1
TT	0

Here, x can take the values 0, 1, 2. Therefore, x is a random variable.

Types of Random Variable:-

1. Discrete Random variable
2. Continuous Random variable

Random variable & Mathematical Expression

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Types of Random Variable:-

1. Discrete Random Variable
2. Continuous Random variable

Discrete Random Variable:- A random variable is called discrete random variable if it can take only isolated values.

Ex- family members, mobile numbers etc.

Continuous Random Variable:- A random variable is called continuous random variable if it can take any values between certain limits.

Ex- Weight, Age etc.

Probability function:-

A function $f(x)$ of a discrete random variable x is called a probability function if it satisfies the following two conditions,

i. $f(x) \geq 0$

ii. $\sum f(x) = 1$

Probability Density function:-

A function $f(x)$ of a continuous random variable x is called a probability density function if it satisfies the following two conditions.

i. $f(x) \geq 0$

ii. $\int_{-\infty}^{\infty} f(x) dx = 1$

Mathematical Expectation:-

If x is a discrete or continuous random variable with probability function or probability density function $f(x)$, then the mathematical expectation of x is usually denoted by $E(x)$ or μ and defined by

$\Rightarrow \mu = E[x] = \sum x f(x)$; If x is a discrete random variable

$\Rightarrow \mu = E[x] = \int_{-\infty}^{\infty} x f(x) dx$; If x is a continuous random variable

Properties of Mathematical Expectation of a random variable:-

1. If " a " is a constant then $E[a] = a$

2. If x is a random variable with expectation $E[x]$ then $E[ax+b] =$

$$E[ax+b] = E[x] + b; \text{ where } a \text{ and } b \text{ is constant.}$$

$$3. E[ax] = aE[x]$$

4. If x and Y are random variables then

$$E[x+Y] = E[x] + E[Y]$$

5. If x and Y are random variables then

$$E[x-Y] = E[x] - E[Y]$$

☐ Properties of variance a random variable:-

1. If 'a' is a constant. then $V[a] = 0$
2. If X is a random variable, then $V[ax+b] = a^2 V[X]$
where a and b are constant.
3. $V(a, X) = a^2 V[X]$
4. $V[X+Y] = V[X] + V[Y]$
5. $V[X-Y] = V[X] - V[Y]$
6. $V[X] = E[X - E(X)]^2 = E(X^2) - [E(X)]^2$

Prb-1

A continuous random variable X has the following probability density function,

$$f(x) = Kx^2, \quad 0 \leq x \leq 1$$

1. Determine the value of K ,
2. $P(X < 0.65)$
3. $P(X > 0.30)$
4. $P(0.25 < X < 0.75)$
5. $E[X]$ or $E[7X+9]$
6. $V[X]$ or $V[7X+9]$
7. $SD(X)$
8. $CV(X)$

Sol:-

1. We know, $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\Rightarrow \int_{0.0}^{1.0} Kx^2 dx = 1$$

$$\Rightarrow K \int_{0.0}^{1.0} x^2 dx = 1$$

$$\Rightarrow K \left[\frac{x^3}{3} \right]_{0.0}^{1.0} = 1$$

$$\Rightarrow \frac{K}{3} (1^3 - 0^3) = 1$$

$$\Rightarrow K = 3$$

Therefore,

$$f(x) = 3x^2; 0 \leq x \leq 1.$$

ii) Probability that x less than 0.65 (Ans).

$$P(x < 0.65) = \int_0^{0.65} 3x^2 dx \quad [K=3]$$

$$= 3 \int_0^{0.65} x^2 dx$$

$$= 3 \left[\frac{x^3}{3} \right]_0^{0.65}$$

$$= (0.65)^3 - 0^3$$

$$= 0.274 \text{ (Ans).}$$

iii) Probability that x greater than 0.30 is $P[x > 0.30]$

$$P(x > 0.30) = \int_{0.30}^1 3x^2 dx \quad [K=3]$$

$$= 3 \int_{0.30}^1 x^2 dx$$

$$= 3 \left[\frac{x^3}{3} \right]_{0.30}^1$$

$$= 1^3 - (0.30)^3$$

$$= 0.973 \text{ (Ans).}$$

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iv. Probability that lies between 0.25 and 0.75

$$P(0.25 < x < 0.75) = \int_{0.25}^{0.75} 3x^2 dx \quad [K=3]$$

$$= 3 \int_{0.25}^{0.75} x^2 dx$$

$$= 3 \left[\frac{x^3}{3} \right]_{0.25}^{0.75}$$

$$= (0.75)^3 - (0.25)^3$$

$$= 0.406.$$

(Ans)

$$v) E[x] = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

$$= \int_0^1 x \cdot 3x^2 dx$$

$$= 3 \int_0^1 x^3 dx$$

$$= 3 \left[\frac{x^4}{4} \right]_0^1$$

$$= \frac{3}{4} [1^4 - 0^4]$$

$$= \frac{3}{4}.$$

$$\therefore E[7x+9] = 7 \cdot E[x] + 9$$

$$= 7 \cdot \frac{3}{4} + 9$$

$$[E[ax+b] = aE[x] + b]$$

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$$= \frac{57}{4} \cdot (\text{Ans}).$$

$$\text{vi) } V(x) = E[x^2] - [E(x)]^2$$

$$= \int_0^1 3x^2 \cdot 3x^2 dx - \left(\frac{3}{4}\right)^2$$

$$= 3 \int_0^1 x^4 dx - \left(\frac{3}{4}\right)^2$$

$$= 3 \left[\frac{x^5}{5} \right]_0^1 - \left(\frac{3}{4}\right)^2$$

$$= \frac{3}{5} (1^5 - 0^5) - \frac{9}{16}$$

$$= \frac{3}{5} - \frac{9}{16}$$

$$= -\frac{3}{80} \cdot (\text{Ans}).$$

$$\text{vii) } V(7x+9) = 7^2 V(x) \because [V(ax+b) = a^2 V(x)]$$

$$= 49 \cdot \frac{3}{80}$$

$$= \frac{147}{80} \cdot (\text{Ans}).$$

viii) We know,

$$SD(x) = \sqrt{V(x)}$$

$$= \sqrt{3/80}$$

$$= \frac{\sqrt{15}}{20} \cdot (\text{Ans}).$$

viii) ~~Q. 11~~

We know,

$$C.V(x) = \frac{SD(x)}{E(x)} \times 100\%$$

$$= \frac{\sqrt{15}/20}{3/4} \times 100\%$$

$$= 25.81\%$$

(Ans).

BORNH

Problem-2

$$f(x) = K(x+1); \quad 2 \leq x \leq 5$$

i. Find the value of K .

ii. $P(x > 3)$

iii. $P(x < 4)$

iv. $E[x]$

v. $V[x]$

vi. $SD[x]$

vii. $C.V[x]$

Probability Distribution

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1. Binomial Distribution (Discrete distribution)
2. Poisson Distribution (Discrete distribution)
3. Normal Distribution (Continuous distribution)

Binomial Distribution:-

A discrete random variable x is said to have binomial distribution if its probability function is as follows

$$f(x) = {}^n C_x p^x \cdot q^{n-x}; \quad x = 0, 1, 2, \dots, n$$

where n and p are the parameters of the distribution and $p + q = 1$.

Here,

n = number of trial

p = probability of success = $\frac{x}{n}$

x = number of success

q = probability of failure.

Assumption of binomial Distribution:-

Four properties of a Binomial Distribution.

1. The sample consists of a fixed number of observations
2. Each observation is classified into one of the two mutually exclusive and collectively exhaustive categories

called success and failure.

3. Probability of an observation being defined as a success p or a failure $1-p$ is constant over all observation.

4. The outcome of any observation is independent of the outcome of any other observation.

☒ Poisson Distribution:-

A discrete random variable x is said to have poisson distribution if its probability, if its probability is as follows,

$$f(x) = \frac{e^{-m} m^x}{x!}, \quad x = 0, 1, 2, \dots, \infty$$

where, m is the parameter of the distribution and $e = 2.718$.

Practical application of poisson distribution:-

1. The number of suicides reported in a particular day.

2. The number of printing mistakes per page of a book.

3. The number of faulty blades in a packet of 100.

4. The number of letters lost in a mail per day.

Normal Distribution:-

A continuous random variable is said to be have normal distribution if its Probability density function is as follows:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-1/2 \left(\frac{x-\mu}{\sigma}\right)^2}; \quad -\infty \leq x \leq \infty$$

$$-\infty \leq \mu \leq \infty$$

$$\sigma > 0$$

where μ and σ are the parameters of the distribution

Importance of Normal distribution:-

1. In practice under certain condition most of the probability and sampling distributions can be approximated by normal distribution.
2. According to certain limit theorem, if mean and variance of distributions can be converted to normal distribution.
3. Normal distribution is the basis of all sampling distribution.
4. Assumption of normality is the basis of all the test of significance in applied statistics.
5. Normal distributions find its application in industrial statistics such as quality control.

Problem-1

A fair coin is tossed 5 times. Find the probability that,

- i. Exactly two heads.
- ii. No head.
- iii. At least three heads.
- iv. At most two heads.

Sol:-

Here, $n=5$.

$$p=q=\frac{1}{2} \text{ [As the coin is fair]}$$

We know,

$$f(x) = {}^n C_x p^x q^{n-x}; x=0,1,2,\dots,n.$$

$$f(x) = {}^5 C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{5-x}; x=0,1,2,\dots,n$$

let,

x be the number of head.

i) Here,

$$x=2.$$

$$\therefore f(x) = {}^5 C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^{5-2}$$

$$= {}^5 C_2 \cdot \frac{1}{4} \cdot \left(\frac{1}{2}\right)^3$$

$$= \frac{5}{54}$$

$$= 0.0926. (\text{Ans}).$$

ii) Here,

$$x=0.$$

$$\therefore f(0) = {}^5C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{5-0}$$

$$= {}^5C_0 \cdot 1 \cdot \left(\frac{1}{2}\right)^5$$

$$= 1 \cdot 1 \cdot \frac{1}{32}$$

$$= \frac{1}{32}$$

(Ans).

iii) At least 3 heads i.e. $x \geq 3$

$$\therefore f(x \geq 3) = f(x=3) + f(x=4) + f(x=5)$$

$$= f(3) + f(4) + f(5)$$

$$= {}^5C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^{5-3} + {}^5C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^{5-4} + {}^5C_5 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^{5-5}$$

$$= 10 \cdot \frac{1}{8} \cdot \frac{1}{4} + 5 \cdot \frac{1}{16} \cdot \frac{1}{2} + 1 \cdot \frac{1}{32} \cdot 1$$

$$= \frac{16}{32}$$

$$= \frac{1}{2} \text{ (Ans).}$$

iv) At most 2 heads, i.e. $x \leq 2$.

$$\begin{aligned}
 P(x \leq 2) &= P(0) + P(1) + P(2) \\
 &= {}^5C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{5-0} + {}^5C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{5-1} + {}^5C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^{5-2} \\
 &= 1 \cdot 1 \cdot \frac{1}{32} + 5 \cdot \frac{1}{2} \cdot \frac{1}{16} + 10 \cdot \frac{1}{4} \cdot \frac{1}{8} \\
 &= \frac{1}{32} + \frac{5}{32} + \frac{5}{32} \\
 &= \frac{11}{32} \text{ (Ans).}
 \end{aligned}$$

Problem-2

A family has 5 children. The probability of being a boy or a girl is equal.

Find the probability that has —

- i. No boys.
- ii. Exactly Two boys.
- iii. At most Two boys.
- iv. At least three boys.

Estimation and Test of Hypothesis

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Hypothesis:-

A statement about the nature of a population

Example: Students who eat breakfast will perform better on a stat exam than students who do not eat breakfast

Test of Hypothesis:-

The statistical procedure which is used to verify any statement or assumption about population parameter on the basis of sample observations is known as test of significance.

Null Hypothesis:- সত্য বিবেচনা research hypothesis

The hypothesis which we are going to test for possible rejection under the assumption that it is true.

Null Hypothesis is denoted by H_0 .

$$H_0 = \mu_1 = \mu_2 \rightarrow \mu = \text{mean}$$

Alternative Hypothesis:-

Each of all possible hypothesis other than null hypothesis is called alternative hypothesis and is usually denoted by H_1

$$H_1, H_0: \mu_1 \neq \mu_2$$

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$\alpha \uparrow, \beta \downarrow$
 $\alpha \downarrow, \beta \uparrow$

☐ Type-I error:- [False Positive]

The error of rejecting H_0 (accepting H_1) which is true is called Type-I error. (α)

☐ Type-II error:- [False negative]

The error of accepting null hypothesis H_0 when it is false is called Type-II error.

Type-II error is denoted by β

☐ Level of significance:-

denoted by α

The probability of Type-I error is called level of significance.

→ It is denoted by α

→ α could be 1%/5%/10%

☐ P value:- Probability value

P value is the least possible value of α for which we can reject the null hypothesis.

$P > \alpha = 5\% \rightarrow$ rejected [H_0]

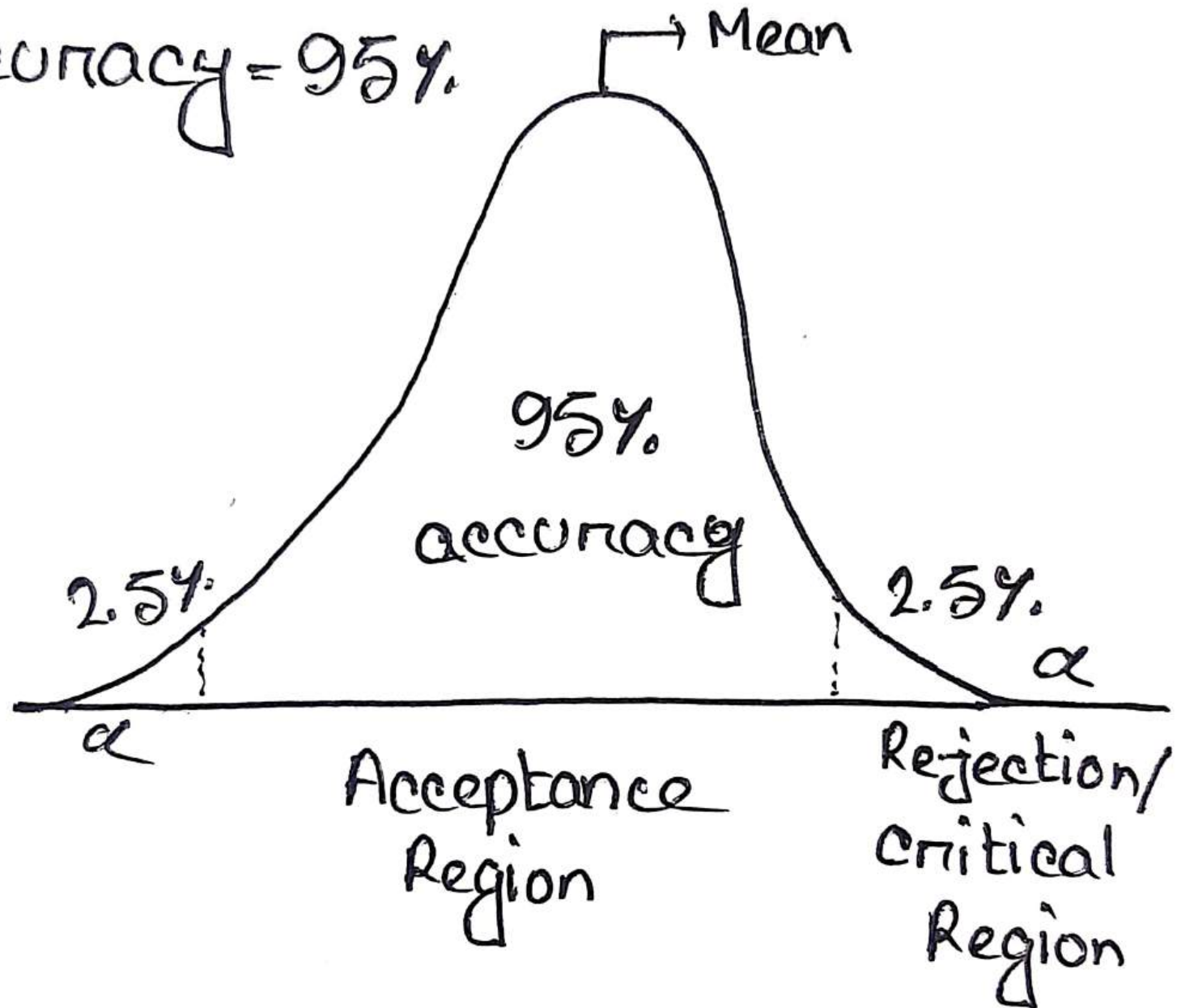
$P < \alpha = 5\% \rightarrow$ accepted [H_0]

df = degree of freedom. → স্বাধীনতার মাত্রা

$$df = (r-1) \times (c-1)$$

$$\alpha = 5\% = 0.05$$

$$\text{accuracy} = 95\%$$



Comment on P value:-

1. If $P < 1\%$; It is highly statistical significant.
2. If $P < 5\%$; It is significant.
3. If $P > 10\%$; It is not significant.

Commonly used Test statistic:-

1. The normal test (Z-test) $\rightarrow$ [large sample size, greater than 30]
2. The 't' test $\rightarrow$ [small sample, less than 30].
3. Chi-square (χ^2) test.
4. F-test.

Application of Chi-square (χ^2) test:-

1. To test the significance of a specified population variance.
2. To test the goodness of fit of a distribution.
3. To test the independence of attributes.
4. To test the equality of several correlation coefficients.
5. To test the homogeneity of several tests.
6. To test the homogeneity of several population variance.

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Application of F-test:-

1. To test the equality of several population means
2. To test the equality of two population variance.
3. To test the significance of regression.
4. To test the significance of multiple correlation coefficient.

Subject / Theme :

Problem-1

Two hundred engineers were interviewed and classified according to their results and job ~~ate~~ satisfaction. The distribution of graduates by results and job satisfaction are given in following contingency table:

Results	Job Satisfaction	
	Yes	No
Excellent	20	70
Good	45	65

Test at 5% level of significance whether there is any relationship between result and job satisfaction.

Note: At 5% level of significance tabulated value of chi-square @ 1df = 3.84.

H_0 : There is no relation between results and job satisfaction.

H_1 : There is relation between results and job satisfaction.

Result	Job satisfaction		Total	
	Yes	No		
Excellent	20(O_{11})	70(O_{12})	90	$\rightarrow R_1$
Good	45(O_{21})	65(O_{22})	110	$\rightarrow R_2$
Total	65	135	200	
	$\downarrow C_1$	$\downarrow C_2$	$\rightarrow N$	

We know,

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

$$E_{ij} = \frac{R_i \times C_j}{N}$$

O = observed value

E = expected value

i = Row & j = col

$$\therefore E_{11} = \frac{90 \times 65}{200} = 29.25$$

$$\therefore E_{12} = \frac{90 \times 135}{200} = 60.75$$

$$\therefore E_{21} = \frac{110 \times 65}{200} = 35.75$$

$$\therefore E_{22} = \frac{110 \times 135}{200} = 74.25$$

$$\begin{aligned} \therefore \chi^2_c &= \frac{(20 - 29.25)^2}{29.25} + \frac{(70 - 60.75)^2}{60.75} + \frac{(45 - 35.75)^2}{35.75} + \frac{(65 - 74.25)^2}{74.25} \\ &= 2.93 + 1.41 + 2.39 + 1.15 \\ &= 7.88. \end{aligned}$$

Here, $\alpha = 5\% = 0.05$ & $df = 1$.

$$\text{also, } \chi^2_c = 3.84$$

$$\chi^2_c > \chi^2_c$$

Comment: H_0 rejected, There is a relationship between results and job satisfaction.